

Chapter 4

Amino Acids and

Proteins

The biological macromolecules are grouped into four classes of molecules that play important roles in cells and in organisms as a whole. All of them are polymers, strings of repeated units [monomers].

This chapter discusses amino acids and proteins from a biological perspective: what they are made of, how they are put together, and what their roles are in the body. These molecules are also discussed in *MCAT Organic Chemistry Review* from an organic chemistry perspective: nomenclature, chirality, etc.

4.1 AMINO ACIDS

Proteins are biological macromolecules that act as enzymes, hormones, receptors, channels, transporters, antibodies, and support structures inside and outside cells. Proteins are composed of 20 different amino acids linked together in polymers. The composition and sequence of amino acids in the polypeptide chain is what makes each protein unique and able to fulfill its special role in the cell. Here, we will start with amino acids, the building blocks of proteins, and work our way up to three-dimensional protein structure and function.

Amino Acid Structure and Nomenclature

Understanding the structure of amino acids is key to understanding both their chemistry and the chemistry of proteins. The generic formula for all 20 amino acids is shown below.

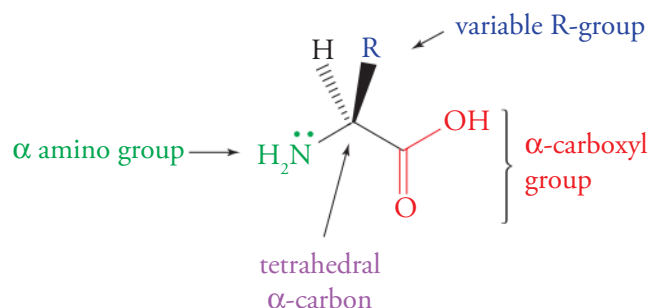


Figure 1 Generic amino acid structure

All 20 amino acids share the same nitrogen-carbon-carbon backbone. The unique feature of each amino acid is its **side chain** (variable R-group), which gives it the physical and chemical properties that distinguish it from the other 19.

Classification of Amino Acids

Each of the 20 amino acids is unique because of its side chain, but many of them are similar in their chemical properties. You should be very familiar with the side chains, and it is important to understand the chemical properties that characterize them, such as their varying *shape*, *ability to hydrogen bond*, and *ability to act as acids or bases (which determines their charge at physiological pH)*.

As you study the 20 amino acids, do so by organizing them into four broad categories: *acidic*, *basic*, *non-polar*, and *polar* amino acids. Each amino acid has a three-letter abbreviation and a one-letter abbreviation, which are both important to know for the MCAT.

Acidic Amino Acids

Aspartic acid and glutamic acid are the only amino acids with carboxylic acid functional groups ($pK_a \approx 4$) in their side chains, thereby making the side chains acidic. Thus, there are three functional groups in these amino acids that may act as acids—the two backbone groups and the R-group. You may hear the terms *aspartate* and *glutamate*—these simply refer to the anionic (deprotonated) form of each molecule, which is how these amino acids are observed at physiological pH.

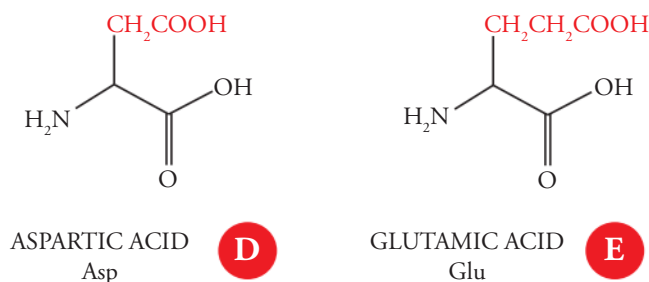


Figure 2 Acidic amino acids

Basic Amino Acids

Lysine, arginine, and histidine have basic R-group side chains. The pK_a values for the side chains in these amino acids are 10 for Lys, 12 for Arg, and 6.5 for His. Both Lys and Arg are cationic (protonated) at physiological pH, but histidine is unique in having a side chain with a pK_a close to physiological pH. At pH 7.4, histidine may be either protonated or deprotonated—we put it in the basic category, but it often acts as an acid too. This makes it a readily available proton acceptor or donor, explaining its prevalence at protein active sites. A mnemonic is “His goes both ways.” This contrasts with amino acids containing $-\text{COOH}$ or $-\text{NH}_2$ side chains, which are *always* anionic (RCOO^-) or cationic (RNH_3^+) at physiological pH. [By the way, *histamine* is a small molecule that has to do with allergic responses, itching, inflammation, and other processes. (You’ve heard of antihistamine drugs, for example.) It is not an amino acid; don’t confuse it with *histidine*.]

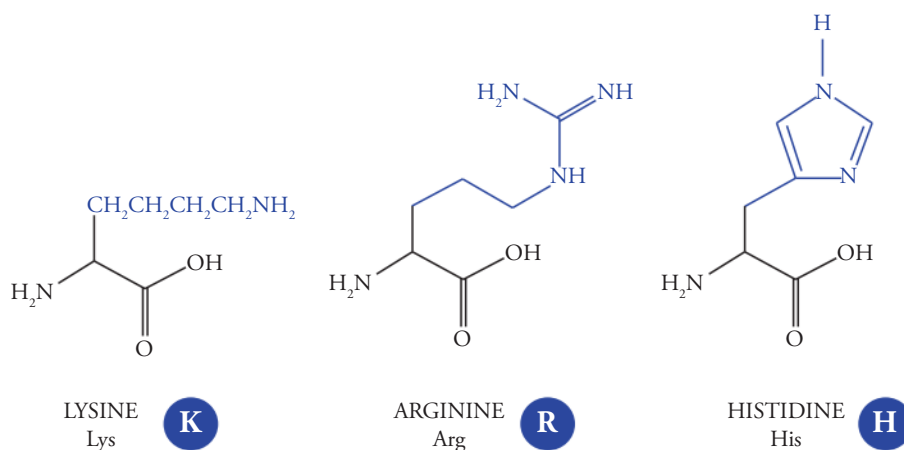


Figure 3 Basic amino acids

Hydrophobic [Nonpolar] Amino Acids

Hydrophobic amino acids have either aliphatic (alkyl) or aromatic side chains. Amino acids with aliphatic side chains include glycine, alanine, valine, leucine, and isoleucine. Amino acids with aromatic side chains include phenylalanine, tryptophan, and tyrosine (though the latter is a polar amino acid). Hydrophobic residues tend to associate with each other rather than with water, and therefore are found on the interior of folded globular proteins, away from water. The larger the hydrophobic group, the greater the hydrophobic force repelling it from water.

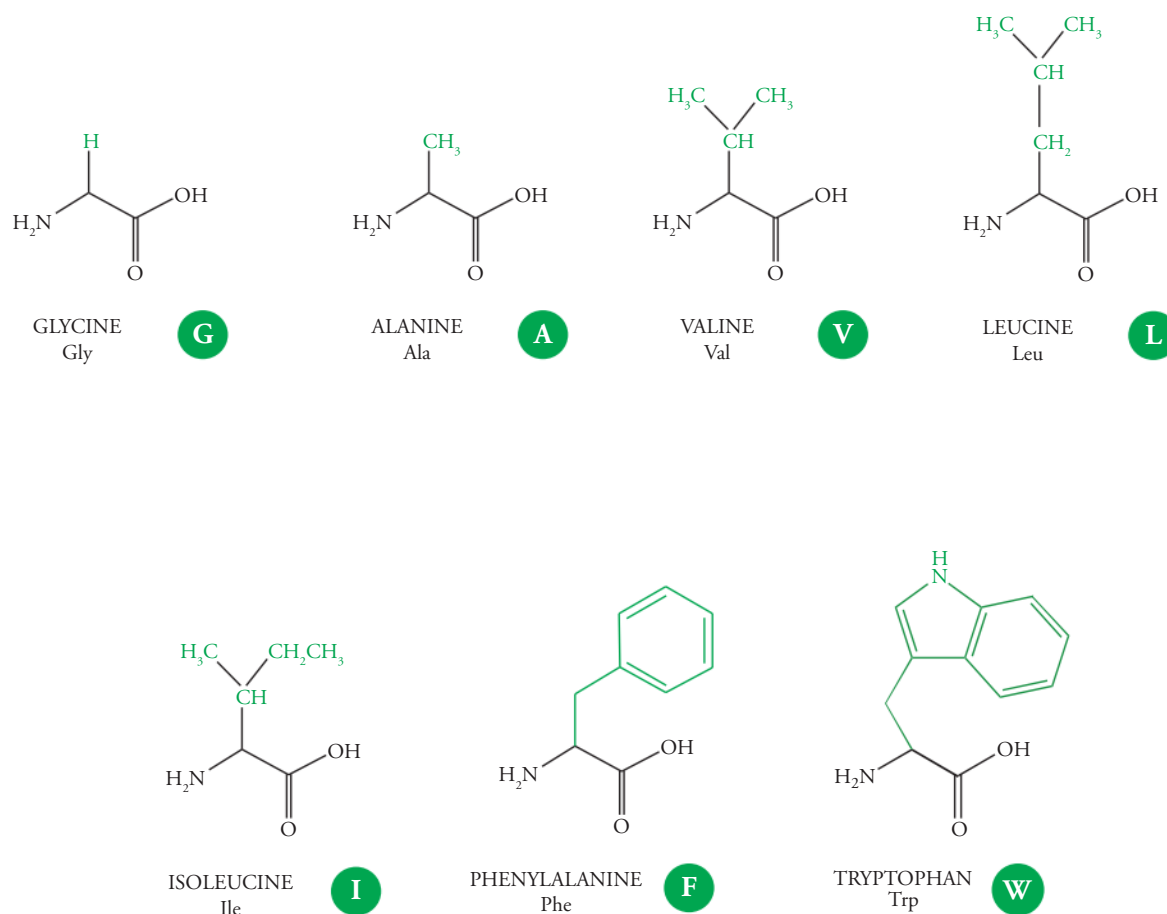


Figure 4 Nonpolar amino acids

Polar Amino Acids

These amino acids are characterized by an R-group that is polar enough to form hydrogen bonds with water but which does not act as an acid or base. This means they are hydrophilic and will interact with water whenever possible. The hydroxyl groups of serine, threonine, and tyrosine residues are often modified by the attachment of a phosphate group by a regulatory enzyme called a kinase. The result is a change

in structure due to the very hydrophilic phosphate group. This modification is an important means of regulating protein activity. This category also includes the amide derivatives of aspartic acid and glutamic acid, which are named asparagine and glutamine, respectively.

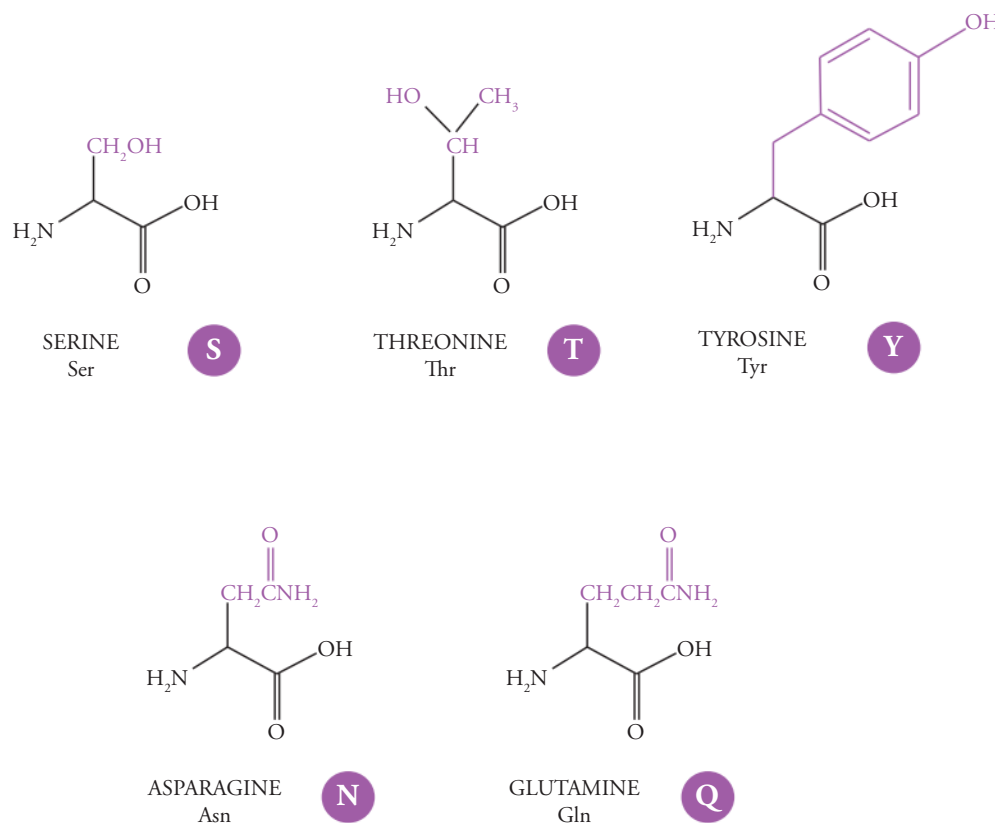


Figure 5 Polar amino acids

Sulfur-Containing Amino Acids

Amino acids with sulfur-containing side chains include cysteine and methionine. Cysteine, which contains a thiol (also called a sulfhydryl—like an alcohol that has an S atom instead of an O atom), is fairly polar, and methionine, which contains a thioether (like an ether that has an S atom instead of an O atom), is fairly nonpolar.

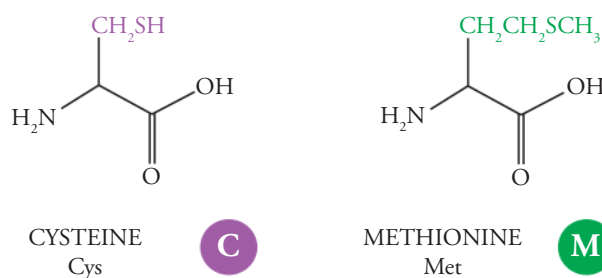
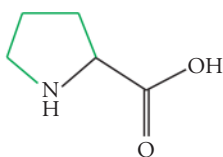


Figure 6 Sulfur-containing amino acids

Proline

Proline is unique among the amino acids in that its amino group is covalently bound to its nonpolar side chain, creating a secondary α -amino group and a distinctive ring structure. This unique feature of proline has important consequences for protein folding (see Section 4.3).



PROLINE
Pro



Table 1 Summary Table of Amino Acids

Hydrophilic			Hydrophobic
ACIDIC	BASIC	POLAR	NONPOLAR
Aspartic acid Glutamic acid	Lysine* Arginine Histidine*	Serine Cysteine Tyrosine Threonine* Asparagine Glutamine	Glycine Alanine Valine* Leucine* Isoleucine* Phenylalanine* Tryptophan* Methionine* Proline
*Denotes one of the nine essential amino acids, those that cannot be synthesized by adult humans and must be obtained from the diet.			

- Which of the following amino acids is most likely to be found on the exterior of a protein at pH 7.0?¹
 - Leucine
 - Alanine
 - Serine
 - Isoleucine

¹ Leucine, alanine, and isoleucine are all hydrophobic residues more likely to be found on the interior than the exterior of proteins. Serine (choice C), which has a hydroxyl group that can hydrogen bond with water, is the correct answer.

4.2 AMINO ACID REACTIVITY

Since amino acids are composed of an acidic group (the carboxylic acid) and a basic group (the amine), we must be sure to understand the acid/base chemistry of amino acids.

Reviewing the Fundamentals of Acid/Base Chemistry

Recall from the previous chapter that amino acids are **amphoteric**, which means that amino acids can act as acids or bases. This should make sense since an amino acid contains the acidic carboxylic acid group and the basic amino group.

Carboxyl groups of amino acids generally have a pK_a of about 2 (stronger acid), while the ammonium groups generally have a pK_a of 9 or 10 (weaker acid).

The mathematical formula that describes the relationship between pH, pK_a , and the position of equilibrium in an acid-base reaction is known as the **Henderson–Hasselbalch** equation:

$$pH = pK_a + \log \frac{[A^-]}{[HA]} = pK_a + \log \frac{[\text{base form}]}{[\text{acid form}]}$$

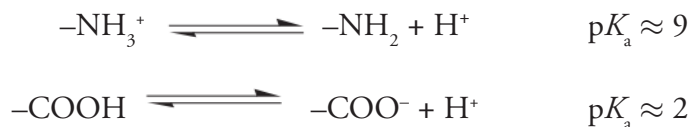
Given the pH and the pK_a , we can calculate the ratio of the base and acid forms of a compound at equilibrium. You will go through this equation much more thoroughly in *MCAT General Chemistry Review*. Don't worry too much about doing any calculations here; just make sure you memorize the following rules:

- When the pH of the solution is less than the pK_a of an acidic group, the acidic group will mostly be in its protonated form.
- When the pH of the solution is greater than the pK_a of an acidic group, the acidic group will mostly be in its deprotonated form.
- Which functional group of amino acids has a stronger tendency to donate protons: carboxyl groups ($pK_a = 2.0$) or ammonium groups ($pK_a = 9$)? Which group will donate protons at the lowest pH?²

² A high pK_a indicates a weak acid. Acids with a low pK_a tend to deprotonate more easily. Therefore, ammonium groups have a stronger tendency to keep their protons, and carboxyl groups will donate protons at the lowest pH (highest $[H^+]$).

Application of Fundamental Acid/Base Chemistry to Amino Acids

All amino acids contain an amino group that acts as a base and a carboxyl group ($pK_a \approx 2$) that acts as an acid. In its protonated, or acidic, form, the amine is called an **ammonium group**, and it has a pK_a between 9 and 10. For example:



- Assuming a pK_a of 2, will a carboxylate group be protonated or deprotonated at pH 1.0?³
- Will the amino group be protonated or deprotonated at pH 1.0?⁴
- Glycine is the simplest amino acid, with only hydrogen as its R-group. Its only functional groups are the backbone groups discussed above (amino and carboxyl). What will be the net charge on a glycine molecule at pH 12?⁵
- At pH 6.0, between the pK_a s of the ammonium and carboxyl groups, what will be the net charge on a molecule of glycine?⁶

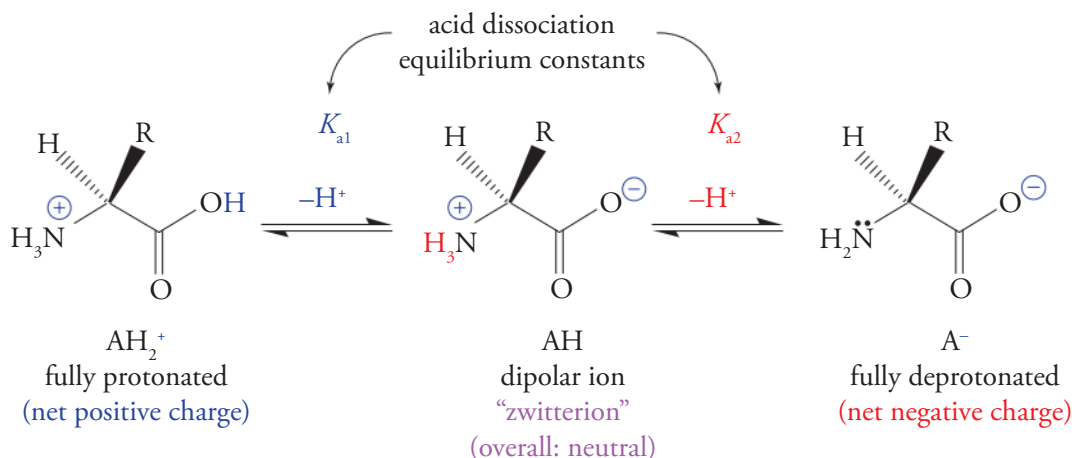


Figure 7 Important amino acid conjugate acid/base pairs

The Isoelectric Point of Amino Acids

There is a pH for every amino acid at which it has no overall net charge (the positive and negative charges cancel). A molecule with positive and negative charges that balance is referred to as a dipolar ion or **zwitterion**. The pH at which a molecule is uncharged (zwitterionic) is referred to as its **isoelectric point** (pI). "Zwitter" is German for "hybrid," implying that an amino acid at its pI has both (+) and (−) charges.

³ The pH is less than the pK_a here, so protonation wins over dissociation, and the group will be protonated. The correct answer is $-\text{COOH}$.

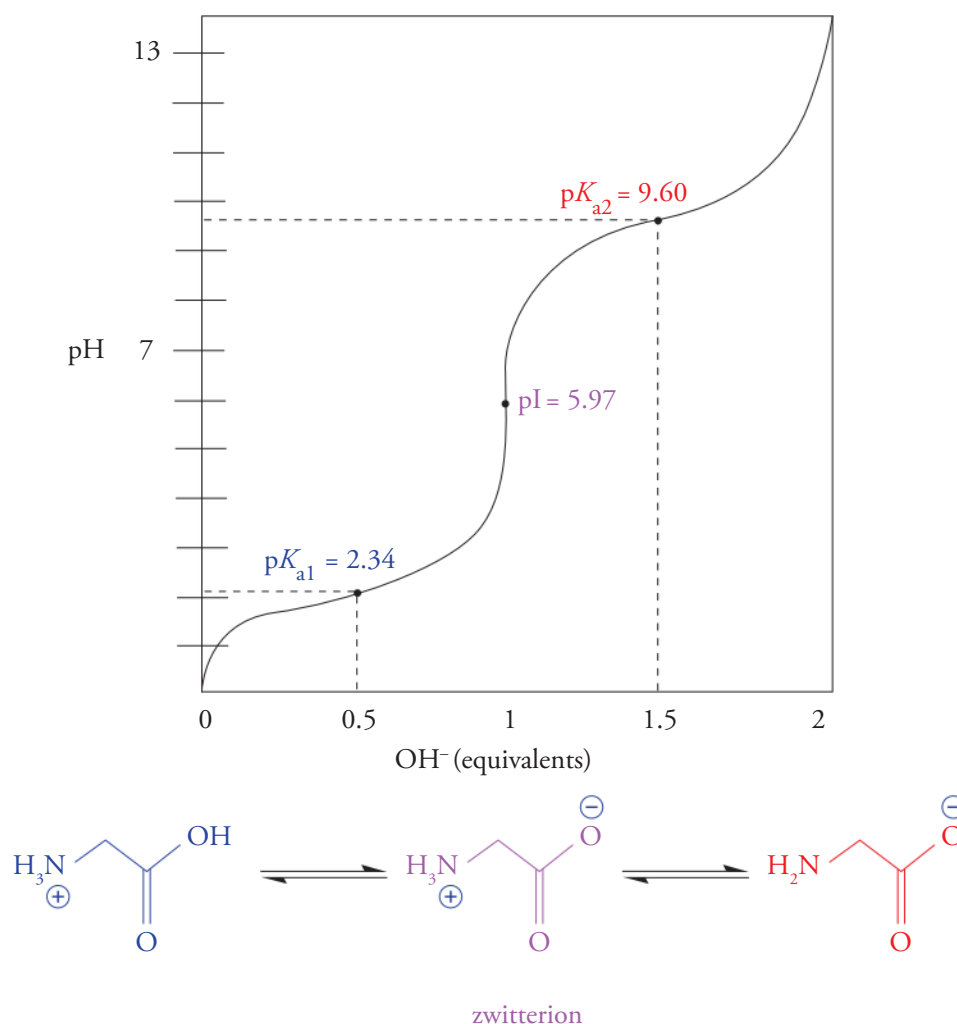
⁴ The pH is much lower than the pK_a for the ammonium group, so the amino group is protonated: NH_3^+ .

⁵ Since pH 12 represents a very low $[\text{H}^+]$, both groups will become deprotonated (COO^- and NH_2), creating a net charge of -1 per glycine molecule.

⁶ The carboxyl group will be deprotonated (COO^-) with a charge of -1 , and the amino group will be protonated (NH_3^+) with a charge of $+1$, creating a net charge of 0 per glycine molecule.

4.2

It is possible to calculate the pI of an amino acid—in other words, to figure out the pH value at which (+) and (−) charges balance (that's the definition of pI). For a molecule with two functional groups, such as glycine, the calculation is simple: just *average the pK_a s of the two functional groups*. The pI of more complex molecules can also be calculated, but the math is complex. For the MCAT, you should know how to calculate the pI of a molecule with two functional groups (with no acidic or basic functional groups in the side chain). Another important thing to know for the MCAT is how to compare the pH of a solution to the pK_a of a functional group of an amino acid and determine if a site is mostly protonated or deprotonated. If the pH is higher than the pK_a , the site is mostly deprotonated; if the pH is lower than the pK_a , the site is mostly protonated. This can be illustrated in the titration curve for glycine (titrations will be discussed in more detail in *MCAT General Chemistry Review*):



- What is the pI of glycine?⁷

⁷ To calculate the pI, just average the pK_a s of the two functional groups: $(9.60 + 2.34)/2 = 5.97$, or roughly 6 (this will be discussed in more detail in *MCAT Organic Chemistry Review*).

4.3 PROTEIN STRUCTURE

There are two common types of covalent bonds between amino acids in proteins: the **peptide bonds** that link amino acids together into polypeptide chains and **disulfide bridges** between cysteine R-groups.

The Peptide Bond

Polypeptides are formed by linking amino acids together in peptide bonds. A peptide bond is formed between the carboxyl group of one amino acid and the α -amino group of another amino acid with the loss of water. The figure below shows the formation of a dipeptide from the amino acids glycine and alanine.

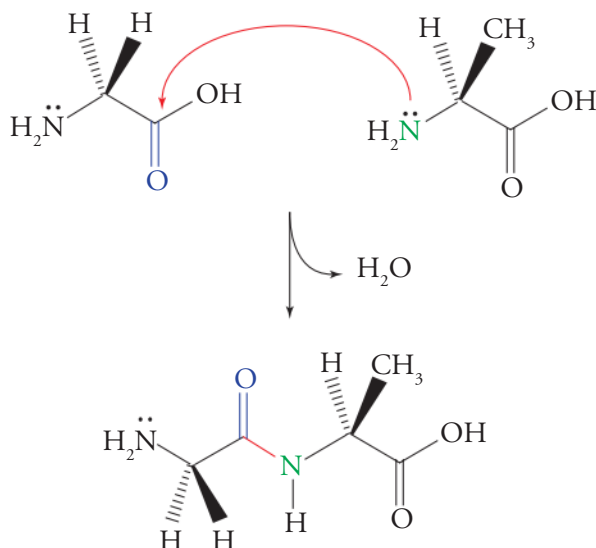


Figure 8 Peptide bond (amide bond) formation

In a polypeptide chain, the N–C–C–N–C–C pattern formed from the amino acids is known as the **backbone** of the polypeptide. An individual amino acid is termed a **residue** when it is part of a polypeptide chain. The amino terminus is the first end made during polypeptide synthesis, and the carboxy terminus is made last. Hence, by convention, the amino-terminal residue is also always written first.

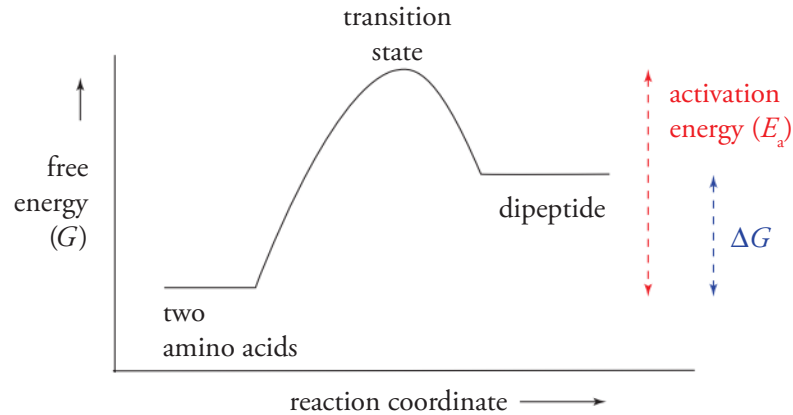
- In the oligopeptide Phe-Glu-Gly-Ser-Ala, state the number of acid and base functional groups, which residue has a free α -amino group, and which residue has a free α -carboxyl group. (Refer to the beginning of the chapter for structures.)⁸
- How many unique dipeptides (made from linking two amino acids) can be synthesized using only alanine and glycine residues?⁹

⁸ As stated above, the amino end is always written first. Therefore, the oligopeptide begins with an exposed Phe amino group and ends with an exposed Ala carboxyl; all the other backbone groups are hitched together in peptide bonds. Out of all the R-groups, there is only one acidic or basic functional group, the acidic glutamate R-group. This R-group plus the two terminal backbone groups gives a total of three acid/base functional groups.

⁹ Four (Gly-Gly, Ala-Ala, Gly-Ala, Ala-Gly). Note that Ala-Gly and Gly-Ala are not identical peptides. In Ala-Gly, the N-terminus is Ala and the C-terminus is Gly. In Gly-Ala, the N-terminus is Gly and the C-terminus is Ala.

4.3

- Thermodynamics states that free energy must decrease for a reaction to proceed spontaneously and that such a reaction will spontaneously move toward equilibrium. The diagram below shows the free energy changes during peptide bond formation. At equilibrium, which is thermodynamically favored: the dipeptide or the individual amino acids?¹⁰



- In that case, how are peptide bonds formed and maintained inside cells?¹¹

Hydrolysis of a protein by another protein is called **proteolysis** or **proteolytic cleavage**, and the protein that does the cutting is known as a **proteolytic enzyme** or **protease**. Proteolytic cleavage is a specific means of cleaving peptide bonds. Many enzymes only cleave the peptide bond adjacent to a specific amino acid. For example, the protease trypsin cleaves on the carboxyl side of the positively charged (basic) residues arginine and lysine, while chymotrypsin cleaves adjacent to large hydrophobic residues such as phenylalanine. (Do *not* memorize these examples.)

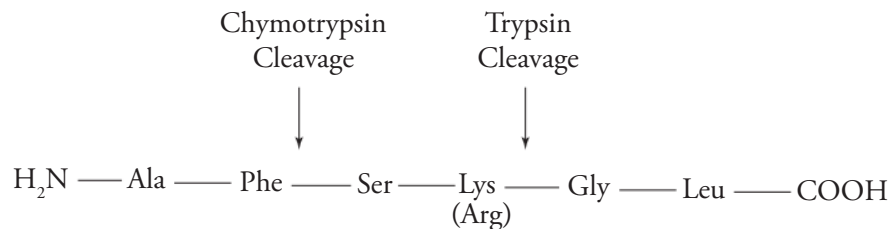


Figure 9 Specificity of protease cleavage

- Based on the above, if the following peptide is cleaved by trypsin, what amino acid will be on the new N-terminus and how many fragments will result: Ala-Gly-Glu-Lys-Phe-Phe-Lys?¹²

¹⁰ The dipeptide has a higher free energy, so its existence is less favorable. In other words, existence of the chain is less favorable than existence of the isolated amino acids.

¹¹ During protein synthesis, stored energy is used to force peptide bonds to form. Once the bond is formed, even though its destruction is thermodynamically favorable, it remains stable because the activation energy for the hydrolysis reaction is so high. In other words, hydrolysis is thermodynamically favorable but kinetically slow.

¹² Trypsin will cleave on the carboxyl side of the Lys residue, with Phe on the N-terminus of the new Phe-Phe-Lys fragment. There will be two fragments after trypsin cleavage: Phe-Phe-Lys and Ala-Gly-Glu-Lys.

The Disulfide Bond

Cysteine is an amino acid with a reactive thiol (sulfhydryl, SH) in its side chain. The thiol of one cysteine can react with the thiol of another cysteine to produce a covalent sulfur-sulfur bond known as a disulfide bond, as illustrated below. The cysteines forming a disulfide bond may be located in the same or different polypeptide chain(s). The disulfide bridge plays an important role in stabilizing tertiary protein structure; this will be discussed in the section on protein folding. Once a cysteine residue becomes disulfide-bonded to another cysteine residue, it is called *cystine* instead of cysteine.

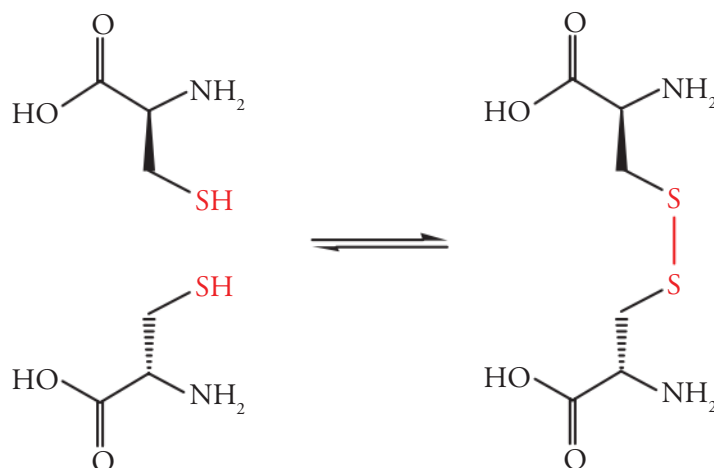


Figure 10 Formation of the disulfide bond

- Which is more oxidized, the sulfur in *cysteine* or the sulfur in *cystine*?¹³
- The inside of cells is known as a reducing environment because cells possess antioxidants (chemicals that prevent oxidation reactions). Where would disulfide bridges be more likely to be found, in extracellular proteins, under oxidizing conditions, or in the interior of cells, in a reducing environment?¹⁴

Protein Structure in Three Dimensions

Each protein folds into a unique three-dimensional structure that is required for that protein to function properly. Improperly folded, or **denatured**, proteins are nonfunctional. There are four levels of protein folding that contribute to their final three-dimensional structure. Each level of structure is dependent upon a particular type of bond, as discussed in the following sections.

¹³ In forming cystine from two cysteine residues, hydrogen atoms are removed (an oxidation reaction), indicating that the sulfur in cystine is more oxidized.

¹⁴ In a reducing environment, the S-S group is reduced to two SH groups. Disulfide bridges are found only in extracellular polypeptides, where they will not be reduced. Examples of protein complexes held together by disulfide bridges include antibodies and the hormone insulin.

4.3

Denaturation is an important concept. It refers to the **disruption of a protein's shape without breaking peptide bonds**. Proteins are denatured by *urea* (which disrupts hydrogen bonding interactions), by *extremes of pH*, by *extremes of temperature*, and by *changes in salt concentration (tonicity)*.

Primary [1°] Structure: The Amino Acid Sequence

The simplest level of protein structure is the order of amino acids bonded to each other in the polypeptide chain. This linear ordering of amino acid residues is known as primary structure. **Primary structure** is the same as **sequence**. The bond which determines 1° structure is the peptide bond, simply because this is the bond that links one amino acid to the next in a polypeptide.

Secondary [2°] Structure: Hydrogen Bonds Between Backbone Groups

Secondary structure refers to the initial folding of a polypeptide chain into shapes stabilized by hydrogen bonds between backbone NH and CO groups. Certain motifs of secondary structure are found in most proteins. The two most common are the α -**helix** and the β -**pleated sheet**.

All α -helices have the same well-defined dimensions that are depicted below with the R-groups omitted for clarity. The α -helices of proteins are always right-handed, 5 angstroms in width, with each subsequent amino acid rising 1.5 angstroms. There are 3.6 amino acid residues per turn with the α -carboxyl oxygen of one amino acid residue hydrogen-bonded to the α -amino proton of an amino acid three residues away. (*Don't* memorize these numbers, but *do* try to visualize what they mean.)

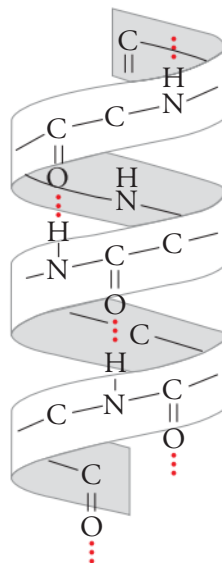


Figure 11 An α -helix

The unique side chain of proline causes two problems in polypeptide chains:

1. The formation of a peptide bond with proline (shown below) eliminates the only hydrogen atom on the nitrogen atom of proline. The absence of the N-H bond disrupts the backbone hydrogen bonding in the polypeptide chain.
2. The unique structure of proline forces it to kink the polypeptide chain.

For both reasons, proline residues never appear within the α -helix.

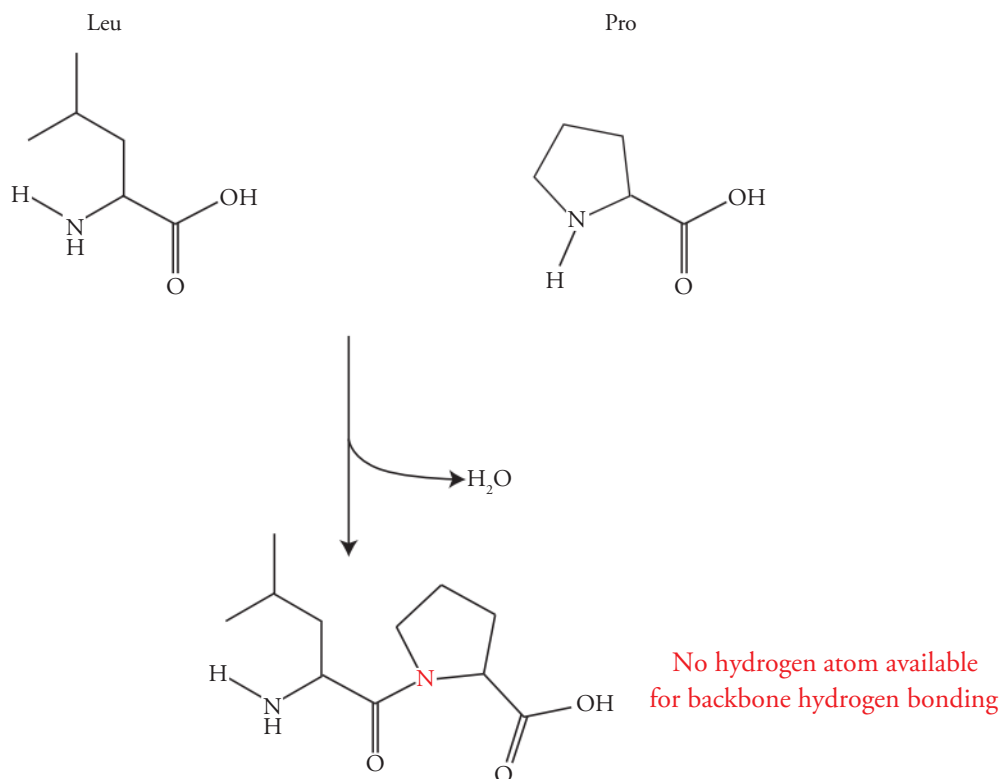


Figure 12 Proline

Proteins such as hormone receptors and ion channels are often found with α -helical transmembrane regions integrated into the hydrophobic membranes of cells. The α -helix is a favorable structure for a hydrophobic transmembrane region because all polar NH and CO groups in the backbone are hydrogen-bonded to each other on the inside of the helix, and thus don't interact with the hydrophobic membrane interior. α -Helical regions that span membranes also have hydrophobic R-groups, which radiate out from the helix, interacting with the hydrophobic interior of the membrane.

β -pleated sheets are also stabilized by hydrogen bonding between NH and CO groups in the polypeptide backbone. In β -sheets, however, hydrogen bonding occurs between residues distant from each other in the chain or even on separate polypeptide chains. Also, the backbone of a β -sheet is extended, rather than coiled, with side groups directed above and below the plane of the β -sheet. There are two types of β -sheets, one with adjacent polypeptide strands running in the *same* direction (**parallel** β -pleated sheet) and another in which the polypeptide strands run in *opposite* directions (**antiparallel** β -pleated sheet).

4.3

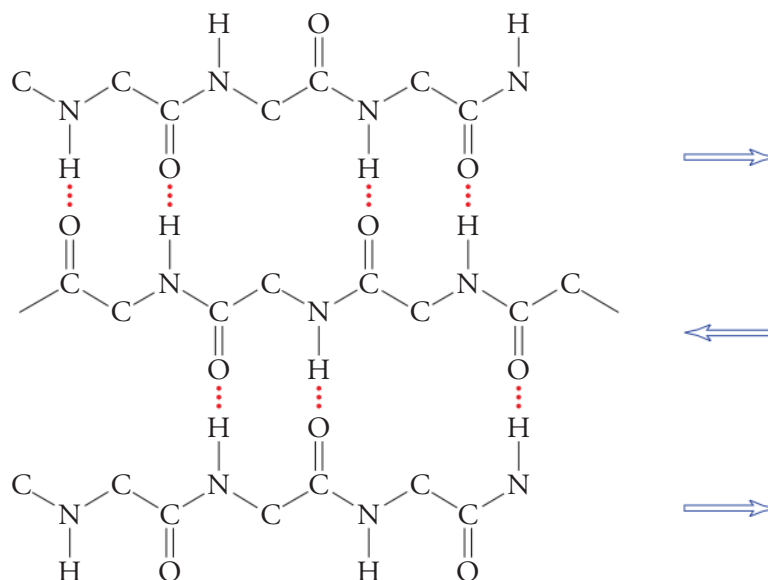


Figure 13 A β -Pleated sheet

- If a single polypeptide folds once and forms a β -pleated sheet with itself, would this be a parallel or an antiparallel β -pleated sheet?¹⁵
- What effect would a molecule that disrupts hydrogen bonding, such as urea, have on protein structure?¹⁶

Tertiary [3°] Structure: Hydrophobic/Hydrophilic Interactions

The next level of protein folding, tertiary structure, concerns interactions between amino acid residues located more distantly from each other in the polypeptide chain. These interactions may include van der Waals forces between nonpolar side chains, hydrogen bonds between polar side chains, disulfide bonds between cysteine residues, and electrostatic interactions between acidic and basic side chains. The folding of secondary structures such as α -helices into higher order tertiary structures is driven by interactions of R-groups with each other and with the solvent (water). Hydrophobic R-groups tend to fold into the interior of the protein, away from the solvent, and hydrophilic R-groups tend to be exposed to water on the surface of the protein (shown for the generic globular protein). This is called the **hydrophobic effect**.

¹⁵ It would be antiparallel because one participant in the β -pleated sheet would have a C to N direction, while the other would be running N to C.

¹⁶ Putting a protein in a urea solution will disrupt H-bonding, thus disrupting secondary structure (and possibly tertiary and quaternary) by unfolding α -helices and β -sheets. It would not affect primary structure, which depends on the much more stable peptide bond. Disruption of 2°, 3°, or 4° structure without breaking peptide bonds is *denaturation*.

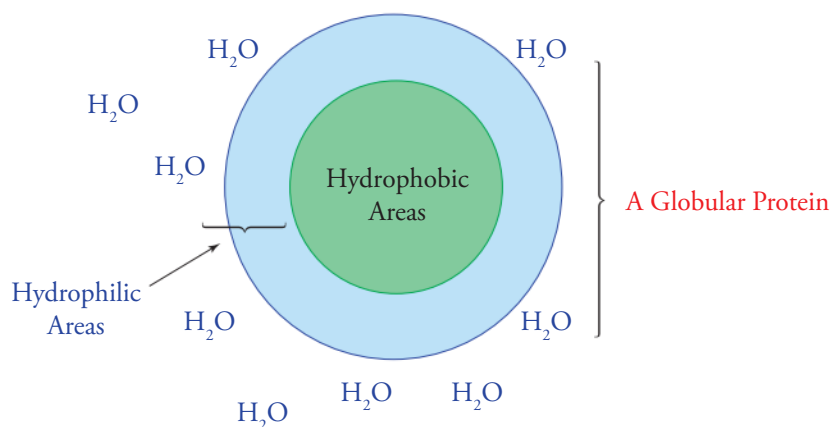


Figure 14 Folding of a globular protein in aqueous solution

Under the right conditions, the forces driving hydrophobic avoidance of water and hydrogen bonding will fold a polypeptide spontaneously into the correct conformation, the lowest energy conformation. In a classic experiment by Christian Anfinsen and coworkers, the effect of a denaturing agent (urea) and a reducing agent (β -mercaptoethanol) on the folding of a protein called ribonuclease was examined. In the following questions, you will reenact their thought processes. Try to answer the questions before reading the footnotes.

- Ribonuclease has eight cysteines that form four disulfide bonds. What effect would a reducing agent have on its tertiary structure?¹⁷
- If the disulfides serve only to lock into place a tertiary protein structure that forms first on its own, then what effect would the reducing agent have on correct protein folding?¹⁸
- Would a protein end up folded normally if you (1) first put it in a reducing environment, (2) then denatured it by adding urea, (3) next removed the reducing agent, allowing disulfide bridges to reform, and (4) finally removed the denaturing agent?¹⁹
- What if you did the same experiment but in this order: 1, 2, 4, 3?²⁰
- Which of the following may be considered an example of tertiary protein structure?²¹
 - I. van der Waals interactions between two Phe R-groups located far apart on a polypeptide
 - II. Hydrogen bonds between backbone amino and carboxyl groups
 - III. Covalent disulfide bonds between cysteine residues located far apart on a polypeptide

¹⁷ The disulfide bridges would be broken. Tertiary structure would be less stable.

¹⁸ The shape should not be disrupted if breaking disulfides is the only disturbance. It's just that the shape would be less sturdy—like a concrete wall without the rebar.

¹⁹ No. If you allow disulfide bridges to form while the protein is still denatured, it will become locked into an abnormal shape.

²⁰ You should end up with the correct structure. In step 1, you break the reinforcing disulfide bridges. In step 2, you denature the protein completely by disrupting H-bonds. In step 4, you allow the H-bonds to reform; as stated in the text, normally the correct tertiary structure will form spontaneously if you leave the polypeptide alone. In step 3, you reform the disulfide bridges, thus locking the structure into its correct form.

²¹ **Item I is true:** this is a good example of 3°. **Item II is false:** this describes 2°, not 3° structure. **Item III is true:** this describes the disulfide bond, which is another example of tertiary structure.

Quaternary [4°] Structure: Various Bonds Between Separate Chains

The highest level of protein structure, quaternary structure, describes interactions between polypeptide subunits. A **subunit** is a single polypeptide chain that is part of a large complex containing many subunits (a **multisubunit complex**). The arrangement of subunits in a multisubunit complex is what we mean by quaternary structure. For example, mammalian RNA polymerase II contains 12 different subunits. The interactions between subunits are instrumental in protein function, as in the cooperative binding of oxygen by each of the four subunits of hemoglobin.

The forces stabilizing quaternary structure are generally the same as those involved in tertiary structure—van der Waals forces, hydrogen bonds, disulfide bonds, and electrostatic interactions. It is key to understand, however, that there is one bond that may not be involved in quaternary structure—the peptide bond—because this bond defines sequence (1° structure).

- What is the difference between a disulfide bridge involved in quaternary structure and one involved in tertiary structure?²²

4.4 PROTEINS AS ENZYMES

Enzymes are biological catalysts. They increase the rate of a reaction by lowering the reaction's activation energy, but they *do not affect* ΔG between reactants and products. As catalysts, enzymes have a kinetic role, *not* a thermodynamic one. Enzymes may alter the rate of a reaction enormously: a reaction that would take a hundred years to reach equilibrium without an enzyme may occur in just seconds with an enzyme.

Given that thousands of enzymes have been discovered, scientists frequently classify them based upon reaction type. Table 2 lists several examples but note that enzymes cannot control the direction in which a reaction proceeds; it is common to see enzymes in a given class function in reverse.

Table 2 Enzyme Classes

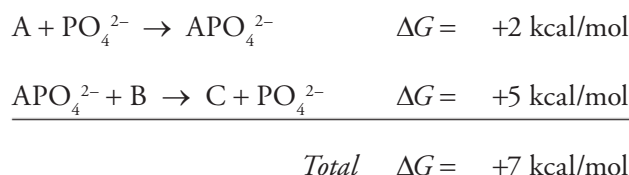
Enzyme Class	Reaction
Hydrolase	hydrolyzes chemical bonds (includes ATPases, proteases, and others)
Isomerase	rearranges bonds within a molecule to form an isomer
Ligase	forms a chemical bond (e.g., DNA ligase)
Lyase	breaks chemical bonds by means other than oxidation or hydrolysis (e.g., pyruvate decarboxylase)
Kinase	transfers a phosphate group to a molecule from a high energy carrier, such as ATP (e.g., phosphofructokinase [PFK])
Oxidoreductase	runs redox reactions (includes oxidases, reductases, dehydrogenases, and others)
Polymerase	polymerization (e.g., addition of nucleotides to the leading strand of DNA by DNA polymerase III)
Phosphatase	removes a phosphate group from a molecule
Phosphorylase	transfers a phosphate group to a molecule from inorganic phosphate (e.g., glycogen phosphorylase)
Protease	hydrolyzes peptide bonds (e.g., trypsin, chymotrypsin, pepsin, etc.)

²² Quaternary disulfides are bonds that form between chains that aren't linked by peptide bonds. Tertiary disulfides are bonds that form between residues in the same polypeptide.

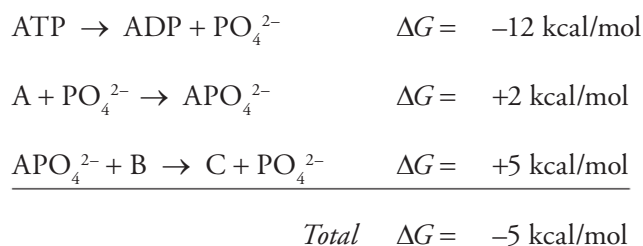
ATP as an Energy Source: Reaction Coupling

Enzymes increase the rate of reactions that have a negative ΔG . These reactions would occur on their own without an enzyme (they are spontaneous) but far more slowly than with one. However, there are many reactions in the body that occur which have a positive ΔG . The biosynthesis of macromolecules such as DNA and protein is not spontaneous ($\Delta G > 0$), but clearly these reactions *do* take place (or we wouldn't be here). How can this be? Thermodynamically unfavorable reactions in the cell can be driven forward by **reaction coupling**. In reaction coupling, one very favorable reaction is used to drive an unfavorable one. This is possible because *free energy changes are additive*. [What is the favorable reaction that the cell can use to drive unfavorable reactions?²³] In the lab, the ΔG° for the hydrolysis of one phosphate group from ATP is -7.3 kcal/mol, so it is a very favorable reaction. In the cell, ΔG is about -12 kcal/mol, so in the cell it is even more favorable. [What's the difference between the situation *in vitro* (lab) under standard conditions and *in vivo* (cell) under nonstandard conditions?²⁴]

How does ATP hydrolysis drive unfavorable reactions? There are many ways. One example is by causing a conformational change in a protein; in this way, ATP hydrolysis can be used to power energy-costly events like transmembrane transport. Another example is by transfer of a phosphate group from ATP to a substrate. Take the unfavorable reaction $A + B \rightarrow C$. Let's say that Reactant A must proceed through an intermediate, APO_4^{2-} , in order to participate. Let's say $\Delta G = +7$ kcal/mol for the overall reaction. What if the two partial reactions have ΔG s as follows:



These reactions will not proceed because the overall ΔG will be $+7$ kcal/mol. What will be the *overall* ΔG if we *couple* the reaction $A + B \rightarrow C$ to the hydrolysis of one ATP? All we have to do is add up all the ΔG values, as follows:



Now the overall reaction, shown below, is thermodynamically favorable. We have *coupled* the unfavorable reaction $A + B \rightarrow C$ to the highly favorable hydrolysis of ATP:



²³ ATP hydrolysis!

²⁴ $Q_{\text{(cell)}} \neq K_{\text{eq}}$. This means that the relative concentrations of ATP and ADP + P_i are not at equilibrium levels in the cell. Actually, $Q_{\text{(cell)}} \ll K_{\text{eq}}$ because the cell keeps a high concentration of ATP around.

4.5

Note that we first stated that the enzyme has only a kinetic role (influencing rate only), not a thermodynamic one (determining favorability). Then we went on to discuss reaction coupling, which allows enzymes to promote otherwise unfavorable reactions. There is no contradiction, however. The only difference is viewing reactions in an isolated manner or in the complex series of linked reactions more commonly found in the body. The same rule applies in either case: ΔG must be negative for either a single reaction or a series of linked reactions to occur spontaneously. In summary:

- One reaction in a test tube—the enzyme is a catalyst with a kinetic role only. It influences the rate of the reaction, but not the outcome.
- Many “real-life” reactions in the cell—enzyme controls outcomes by selectively promoting unfavorable reactions via reaction coupling.

4.5 ENZYME STRUCTURE AND FUNCTION

Most enzymes are proteins that must fold into specific three-dimensional structures to act as catalysts. (Some enzymes are RNA or contain RNA sequences with catalytic activity. Most catalyze their own splicing, and the rRNA in ribosomes helps in peptide-bond formation.) An enzyme may consist of a single polypeptide chain or several polypeptide subunits held together in a _____²⁵ (primary? secondary? etc.) structure. The reason for the importance of folding in enzyme function is the proper formation of the **active site**, the region in an enzyme’s three-dimensional structure that is directly involved in catalysis. [What shape are enzymes more likely to have: fibrous/elongated or globular/spherical?²⁶] The reactants in an enzyme-catalyzed reaction are called **substrates**. (Products have no special name; they’re just “products.”) What is the role of the active site, that is, how do enzymes work? The **active site model**, commonly referred to as the “lock and key hypothesis,” states that the substrate and active site are perfectly complementary. This differs from the **induced fit model**, which asserts that the substrate and active site differ slightly in structure and that the binding of the substrate induces a conformational change in the enzyme. The induced fit model has gained greater acceptance in recent years, but regardless of the model, enzymes accelerate the rate of a given reaction by helping to *stabilize the transition state*. For example, if a transition state intermediate possesses a transient negative charge, what amino acid residues might be found at the active site to stabilize the transition state?²⁷ This lowers the activation energy barrier between reactants and products. In our previous example of Bob looking for a job, the use of a career planning service would function as an enzyme by making the process of job hunting easier.

- Is it possible that amino acids located far apart from each other in the primary protein sequence may play a role in the formation of the same active site?²⁸
- If, during an enzyme-catalyzed reaction, an intermediate forms in which the substrate is covalently linked to the enzyme via a serine residue, can this occur at any serine residue or must it occur at a specific serine residue?²⁹

²⁵ quaternary

²⁶ Globular. Structural proteins such as collagen tend to be fibrous, but proteins that act as catalysts tend to be roughly spherical to form an active site in a cleft in the sphere.

²⁷ A positive charge would stabilize the negative charge in the intermediate. Such a charge might be contributed by His, Arg, or Lys. Alternatively, the hydrogen of the $-\text{NH}_2$ group in glutamine or asparagine could hydrogen bond with the negative charge.

²⁸ Yes, the amino acids at the active site may be distant from each other in a polypeptide’s primary sequence but be near each other in the final folded protein. This is why protein folding is crucial for enzyme function.

²⁹ It must occur at a particular serine residue which sticks out into the active site.

- Compound A converts into Compound B in solution: $A \rightleftharpoons B$. The reaction has the following equilibrium constant: $K_{eq} = [B]_{eq} / [A]_{eq} = 1,000$. If pure A is dissolved in water at 298 K, will ΔG for the reaction $A \rightleftharpoons B$ be positive or negative? Is it possible to answer this question without knowing ΔG° ?³⁰
- Regarding the reaction described in the previous question, if pure B is put into solution in the presence of an enzyme that catalyzes the reaction between A and B, which one of the following will be true?³¹
 - All the B will be converted into A, until there is 1,000 times more A than B.
 - All of the B will remain as B, since B is favored at equilibrium.
 - The enzyme will have no effect, since enzymes act on the transition state and there is no transition state present.
 - The reaction that produces A will predominate until $\Delta G = 0$.

The active site for enzymes is generally highly specific in its substrate recognition, including stereospecificity (the ability to distinguish between stereoisomers). For example, enzymes which catalyze reactions involving amino acids are specific for D or L amino acids, and enzymes catalyzing reactions involving monosaccharides may distinguish between stereoisomers as well. [Which configurations are found in animals?³²]

Many **proteases** (protein-cleaving enzymes) have an active site with a serine residue whose OH group can act as a nucleophile, attacking the carbonyl carbon of an amino acid residue in a polypeptide chain. Examples are trypsin, chymotrypsin, and elastase. These enzymes also usually have a **recognition pocket** near the active site. This is a pocket in the enzyme's structure which attracts certain residues on substrate polypeptides. The enzyme always cuts polypeptides at the same site, just to one side of the recognition residue. For example, chymotrypsin always cuts on the carboxyl side of one of the large hydrophobic or aromatic residues Tyr, Trp, Phe, and Met. Enzymes that act on hydrophobic substrates have hydrophobic amino acids in their active sites, while hydrophilic/polar amino acids will comprise the active site of enzymes with hydrophilic substrates.

Given the importance of the active site, it becomes clear that small alterations in its structure can drastically alter enzymatic activity. Therefore, both temperature and pH play a critical role in enzymatic function. As temperature increases, the thermal motion of the peptide and surrounding solution destabilize its structure. If the temperature rises sufficiently, the protein **denatures** and loses its orderly structure. The pH of the surrounding medium also impacts protein stability; several amino acids possess ionizable $-R$ groups that change charge depending on pH. This can decrease the affinity of a substrate for the active site and, if the pH deviates sufficiently, the protein can denature.

³⁰ You don't need to calculate ΔG° ; all you need to know is that with a K_{eq} of 1,000, there will be 1,000 times more B than A in solution at equilibrium. If we create a solution with only A, the reaction must move spontaneously toward B.

³¹ If only B exists in solution, then the back-reaction producing A will predominate until equilibrium is reached ($\Delta G = 0$), regardless of the presence or absence of enzyme (choice **D** is correct, and choice B is wrong). According to the K_{eq} given, at equilibrium there will be 1,000 times more B than A, not the other way around (choice A is wrong). Note that enzymes do not act on the transition state; they act to produce the transition state (choice C is wrong).

³² L amino acids and D sugars. Remember the L in aLaanine.

4.6

- The transition state for a reaction possesses a transient negative charge. The active site for an enzyme catalyzing this reaction contains a His residue to stabilize the intermediate. If the His residue at the active site is replaced by a glutamate that is negatively charged at pH 7.0, what effect will this have on the reaction, assuming that the reactants are present in excess compared to the enzyme?³³
 - A) The repulsion caused by the negative charge in the glutamate at the altered active site will increase the activation energy and make the reaction proceed more slowly than it would in a solution without enzyme.
 - B) The rate of catalysis will be unaffected, but the equilibrium ratio of products and reactants will change, favoring reactants.
 - C) The transition state intermediate will not be stabilized as effectively by the altered enzyme, lowering the rate relative to the rate with catalysis by the normal enzyme.
 - D) The rate of catalysis will decrease, and the equilibrium constant will change.

Enzymatic function can also depend upon the association of additional molecules. **Cofactors**, which are metal ions or small molecules (not themselves a protein), are required for activity in many enzymes. In fact, the majority of the vitamins in our diet serve as precursors for cofactors (e.g., niacin [B3] is ultimately transformed into NAD⁺). When a cofactor is an organic molecule, it is referred to as a **coenzyme**; these often bind to the substrate during the catalyzed reaction. One prime example of a coenzyme, which we will focus on later in the chapter, is coenzyme A (CoA).

4.6 REGULATION OF ENZYME ACTIVITY

Metabolic pathways in the cell are not all continually on, but must be tightly regulated to maintain health. For example, if glycogen synthesis and breakdown occur in the same cell at the same time, a great deal of energy will be wasted without accomplishing anything. Therefore, the activity of key enzymes in metabolic pathways is usually regulated in one or more of the following ways:

- 1) **Covalent modification.** Proteins can have several different groups covalently attached to them, and this can regulate their activity, lifespan in the cell, and/or cellular location. The addition of a phosphoryl group from a molecule of ATP by a protein **kinase** to the hydroxyl of serine, threonine, or tyrosine residues is the most common example. Phosphorylation of these different sites on an enzyme can either activate or inactivate the enzyme. Protein **phosphorylases** also phosphorylate proteins, but use free-floating inorganic phosphate (P_i) in the cell instead of ATP. Protein phosphorylation can be reversed by protein **phosphatases**.
- 2) **Proteolytic cleavage.** Many enzymes (and other proteins) are synthesized in inactive forms (zymogens) that are activated by cleavage by a protease.

³³ Beware of long, complex-sounding questions! They may not be as bad as they look; for instance, the phrase “assuming that the reactants are present in excess compared to the enzyme” adds nothing to the substance of this question. If His (which is positive or neutral at pH 7) is replaced by Glu (negatively charged at pH 7), this could decrease the effectiveness of—or destroy altogether—the active site of the enzyme. This means the transition state would not be effectively stabilized, and the rate of the reaction would simply reduce to that of the uncatalyzed reaction (choice C is correct). The rate would not proceed more slowly than the uncatalyzed reaction (i.e., “in solution without enzyme”; choice A is wrong), and remember that enzymes do not alter reaction equilibria (K_{eq} will be unaffected; choices B and D are wrong).

- 3) **Association with other polypeptides.** Some enzymes have catalytic activity in one polypeptide subunit that is regulated by association with a separate regulatory subunit. For example, there are some proteins that demonstrate continuous rapid catalysis if their regulatory subunit is removed; this is known as **constitutive activity** (*constitutive* means continuous or unregulated). There are other proteins that require association with another peptide in order to function. Still other proteins can bind many regulatory subunits. There are numerous examples of this in the cell, and many of them have diverse and complex regulatory mechanisms that all revolve around the theme of “associations with other polypeptides can affect enzyme activity.”
- 4) **Allosteric regulation.** This is the modification of active-site activity through interactions of molecules with other specific sites on the enzyme (called **allosteric sites**). Let’s look at this in a little more detail.

4.6

Allosteric Regulation

If the cell is to make use of the enzyme as a biochemical switch, there must be a way to turn the enzyme *on* or *off*. One mechanism of regulation is the binding of small molecules to particular sites on an enzyme that are distinct from the active site; this is allosteric regulation. This name comes from the fact that the particular spot on the enzyme which can bind the small molecule is *not* the active site; *allo* means “other,” and *steric* refers to a location in space (as in “steric hindrance”), so *allosteric* means “at another place.” The binding of the allosteric regulator to the allosteric site is generally noncovalent and reversible. When bound, the allosteric regulator can alter the conformation of the enzyme to increase or decrease catalysis, even though it may be bound to the enzyme at a site distant from the active site or even on a separate polypeptide.

Feedback Inhibition

Enzymes usually act as part of pathways, not alone. Rather than regulate every enzyme in a pathway, usually there are one or two key enzymes that are regulated, such as the enzyme that catalyzes the first irreversible step in a pathway. The easiest way to explain this is with an example. Three enzymes (E1, E2, and E3) catalyze the three steps required to convert Substrate A to Product D. When plenty of D is around, it would be logical to shut off E1 so that excess B, C, and D are not made. This would conserve A and would also conserve energy. Commonly, an end-product such as D will shut off an enzyme early in the pathway, such as E1. This is called **negative feedback**, or **feedback inhibition**.

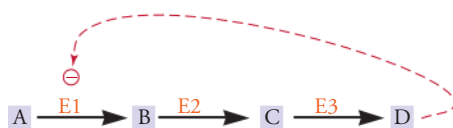


Figure 15 Feedback inhibition

There are examples of positive feedback (“feedback *stimulation*”), but negative feedback is by far the most common example of feedback regulation. On the other hand, *feedforward stimulation* is common. This involves the stimulation of an enzyme by its substrate or by a molecule used in the synthesis of the substrate. For example, in Figure 15, A might stimulate E3. This makes sense because when lots of A is around, we want the pathway for utilization of A to be active.

Allosteric regulation can be quite complex. It is possible for more than one small molecule to be capable of binding to an allosteric site. For example, imagine a reaction pathway from A through Z, in which each step ($A \rightarrow B$, $B \rightarrow C$, etc.) is catalyzed by an enzyme. Let’s say that an allosteric enzyme called E15 catalyzes the reaction $O \rightarrow P$. It would be possible for A to allosterically activate E15 (feedforward stimulation) and for Z to allosterically inhibit E15 (feedback inhibition). This may sound complex, but it’s quite logical. What it means is that when lots of A is around, E15 will be stimulated to use the molecules made from A (B, C, D, etc.) to make P, which could then be used to make Q, R, S, etc., all the way up to Z. On the other hand, if a lot of excess Z built up, it would inhibit E15, thereby conserving the supply of A, B, C, etc. and preventing more build-up of Z, Y, X, etc. Therefore, in addition to acting as switches, enzymes act as *valves*, because they regulate the flow of substrates into products.

4.7

4.7 ENZYME KINETICS

Enzyme kinetics is the study of the rate of formation of products from substrates in the presence of an enzyme. The **reaction rate** (V , for velocity) is the amount of product formed per unit time, in moles per second (mol/s). It depends on the concentration of substrate, $[S]$, and enzyme.³⁴ If there is only a little substrate, then the rate V is directly proportional to the amount of substrate added: double the amount of substrate and the reaction rate doubles, triple the substrate and the rate triples, and so forth. But eventually there is so much substrate that the active sites of the enzymes are occupied much of the time, and adding more substrate doesn’t increase the reaction rate as much, that is, the slope of the V vs. $[S]$ curve decreases. Finally, there is so much substrate that every active site is continuously occupied, and adding more substrate doesn’t increase the reaction rate at all. At this point, the enzyme is said to be **saturated**. The reaction rate when the enzyme is saturated is denoted $V_{\max}$ (see Figure 16). This is a property of each enzyme at a particular concentration of enzyme. You can look it up in a book for the common ones. [If a small amount of enzyme in a solution is acting at $V_{\max}$ and the substrate concentration is doubled, what is the new reaction rate?³⁵]

Another commonly used parameter on these enzyme kinetics graphs is the Michaelis constant K_m . K_m is the substrate concentration at which the reaction velocity is half its maximum. To find K_m on the enzyme kinetics graph, mark the $V_{\max}$ on the y -axis; then divide this distance in half to find $V_{\max}/2$. K_m is found by drawing a horizontal line from $V_{\max}/2$ to the curve and then a vertical line down to the x -axis. K_m is unique for each enzyme-substrate pair and gives information on the affinity of the enzyme for its substrate. If an enzyme-substrate pair has a low K_m , it means that not very much substrate is required to get the reaction rate to half the maximum rate; thus, the enzyme has a high affinity for this particular substrate.

³⁴ Usually the concentration of enzyme is kept fixed, and $[S]$ is taken as the only independent variable (the one the rate depends on). This is applicable to biological systems, in which substrate concentrations change much more than enzyme concentrations.

³⁵ If the enzyme is acting at $V_{\max}$, it is saturated with substrate; adding more substrate will not increase the reaction rate; the rate is still $V_{\max}$.

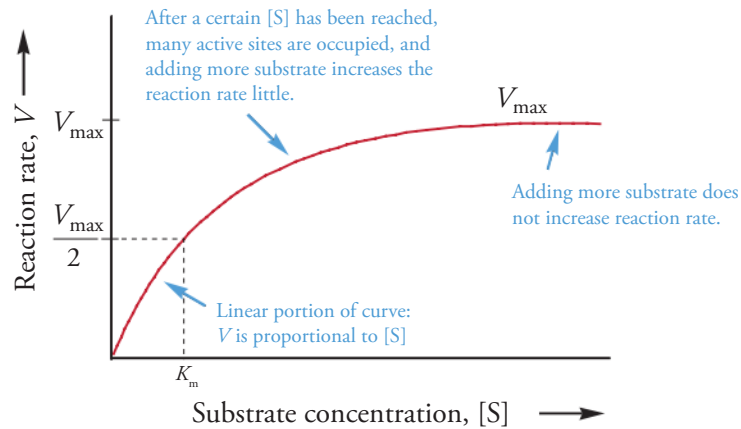


Figure 16 Saturation kinetics

4.7

Cooperativity

Many multisubunit enzymes do not behave in the simple kinetic manner described above. In such enzymes, the binding of substrate to one subunit modulates the affinity of other subunits for substrate. Such enzymes are said to bind substrate cooperatively. There are two types of cooperativity: positive and negative. In positive cooperativity, the binding of a substrate to one subunit increases the affinity of the other subunits for substrate. The conformation of the enzyme prior to substrate binding, with low substrate affinity, is sometimes termed “tense,” and the conformation of enzyme with increased affinity is termed “relaxed”³⁶ (Figure 17). Negative cooperativity (which is less important for the MCAT) is the opposite: the binding of a substrate to one subunit reduces the affinity of the other subunits for substrate. Cooperative enzymes must have more than one active site. They are usually multisubunit complexes, composed of more than one protein chain held together in a quaternary structure. They may also be a single-subunit enzyme with two or more active sites.

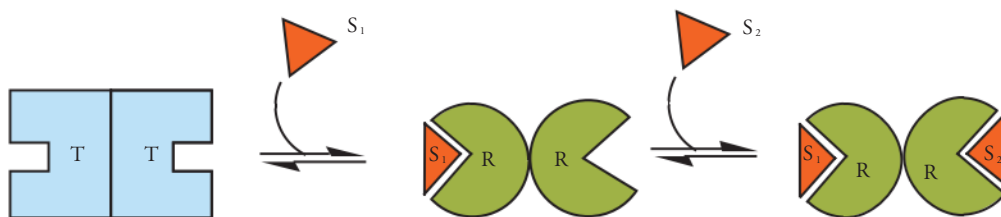


Figure 17 Positive enzyme cooperativity

³⁶ Imagine a group of people who can't get any dates. They are all depressed about it, and they keep each other depressed, which makes it even less likely that any will get a date. They are tense, “turned off,” and inactive. Then one of the depressed group gets a date and gets so excited about it that all the other friends in the group get so enthusiastic that they get dates too. They are “turned on,” relaxed, hip, groovy, and active.

A sigmoidal curve results from positive cooperative binding. In Figure 18 below, the flat part at the bottom left (Region 1) is explained by the notion that at low $[S]$ the enzyme complex has a low affinity for substrate (is in the tense state), and adding more substrate increases the rate little. The steep part in the middle of the curve (Region 2) represents the range of substrate concentrations in which adding substrate greatly increases the reaction rate, because the enzyme complex is in the relaxed state. [The leveling off at the upper right part of the curve (Region 3) represents what?³⁷]

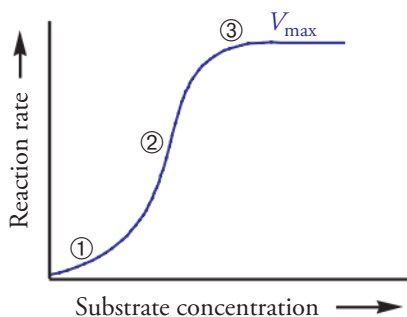


Figure 18 Sigmoidal kinetics of positive cooperativity

Cooperativity does not apply just to catalytic enzymes. For example, hemoglobin (Hb) is a protein complex made of four polypeptide subunits, each of which contains a heme prosthetic group with a single O_2 -binding site. (So one Hb has four hemes and four binding sites.) Hb is a carrier (of oxygen), not a catalyst of any reaction (not an enzyme). It exhibits positively cooperative O_2 binding. This is why the Hb- O_2 dissociation curve is sigmoidal. [What is the relationship between the two notions *allosteric* and *cooperative*?³⁸]

³⁷ Saturation, just as in the case of a noncooperative enzyme.

³⁸ Cooperativity is a special kind of allosteric interaction. One active site acts like an allosteric regulatory site for the other active sites. Secondly, cooperative enzyme complexes are often allosterically regulated also. Hb is an excellent example. Not only does O_2 binding to one subunit increase the other subunits' affinities, but also several other molecules can bind to various sites to change the affinity of the complex. For example, CO_2 stabilizes tense Hb, causing each of the four binding sites to have a lower affinity for oxygen. As a result, in the presence of CO_2 , Hb tends to give up whatever O_2 it has bound. The most important thing to remember, though, is that the binding in cooperativity takes place at the active site, while the binding in allosteric regulation takes place at "other sites."

4.8 INHIBITION OF ENZYME ACTIVITY

Enzyme inhibitors can reduce enzyme activity by a few different mechanisms, including **competitive inhibition**, **noncompetitive inhibition**, **uncompetitive inhibition**, and **mixed-type inhibition**. **Competitive inhibitors** are molecules that *compete* with substrate for binding at the active site. [You can predict that structurally, competitive inhibitors resemble what?³⁹] The key thing to remember about competitive inhibitors is that their inhibition can be overcome by adding more substrate; if the substrate concentration is high enough, the substrate can *outcompete* the inhibitor. Hence, $V_{\max}$ is not affected. You can get to the same $V_{\max}$, but it takes more substrate (see Figure 19). Therefore, the K_m of the reaction to which a competitive inhibitor has been added is increased compared to the K_m of the uninhibited reaction. [If an enzyme has a reaction rate of 1 $\mu\text{mole}/\text{min}$ at a substrate concentration of 50 μM and a rate of 10 $\mu\text{mole}/\text{min}$ at a substrate concentration of 100 μM , does this indicate the presence of a competitive inhibitor?⁴⁰]

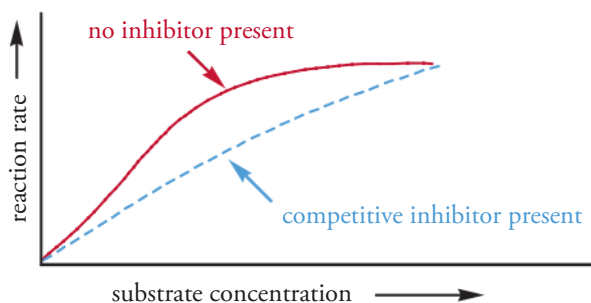


Figure 19 Competitive inhibition

Noncompetitive inhibitors bind at an allosteric site, not at the active site. No matter how much substrate you add, the inhibitor will not be displaced from its site of action (see Figure 20). Hence, noncompetitive inhibition *does* diminish $V_{\max}$. Remember that $V_{\max}$ is always calculated at the same enzyme concentration, since adding more enzyme will increase the measured $V_{\max}$. Addition of a noncompetitive inhibitor changes the $V_{\max}$ and $V_{\max}/2$ of the reaction, but typically does not alter K_m . This is because the substrate can still bind to the active site, but the inhibitor prevents the catalytic activity of the enzyme.

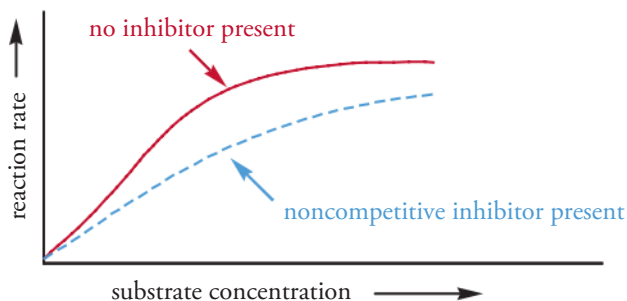
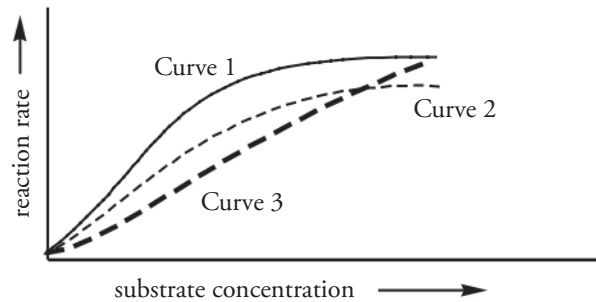


Figure 20 Noncompetitive inhibition

³⁹ Structurally, competitive inhibitors must at least resemble the substrate; however, the most effective competitive inhibitors resemble the transition state that the active site normally stabilizes.

⁴⁰ No. The rate increase is greater than linear, indicating that the effect is caused by cooperativity.

- Carbon dioxide is an allosteric inhibitor of hemoglobin. It dissociates easily when Hb passes through the lungs, where the CO_2 can be exhaled. Carbon *monoxide*, on the other hand, binds at the oxygen-binding site with an affinity 300 times greater than oxygen; it can be displaced by oxygen, but only when there is much more O_2 than CO in the environment. Which of the following is/are correct?⁴¹
 - Carbon monoxide is an irreversible inhibitor.
 - CO_2 is a reversible inhibitor.
 - CO_2 is a noncompetitive inhibitor.
- In the figure below, the kinetics of an enzyme are plotted. In each case, an inhibitor may be present or absent. Which one of the following statements is true?⁴²



- Curve 3 represents noncompetitive inhibition of the enzyme.
- Curve 1 represents noncompetitive inhibition of the enzyme.
- The $V_{\max}$ values of Curve 2 and Curve 3 are the same.
- Curve 3 represents competitive inhibition of the enzyme, and the enzyme is uninhibited in Curve 1.

If an inhibitor is only able to bind to the enzyme-substrate complex (that is, it cannot bind before the substrate has bound), it is referred to as an **uncompetitive inhibitor**. Uncompetitive inhibitors, like noncompetitive inhibitors, bind to allosteric sites. This effectively decreases $V_{\max}$ by limiting the amount of available enzyme-substrate complex which can be converted to product. By sequestering enzyme bound to substrate, this increases the apparent affinity of the enzyme for the substrate as it cannot readily dissociate (decreasing K_m).

⁴¹ Item I: False. The question states that CO can be displaced by oxygen. **Item II: True.** The question states that it dissociates easily. **Item III: True.** The question states it binds allosterically, which means “at another site” (not the active site).

⁴² Since Curve 3 and Curve 1 have the same $V_{\max}$, but Curve 3 has a reduced rate of product formation, it suggests that Curve 3 represents competitive inhibition of the enzyme in Curve 1 (choice D is correct). If Curve 3 represented noncompetitive inhibition, its $V_{\max}$ would be reduced compared to Curve 1 (choice A is wrong), and in no case would an inhibitor have a higher $V_{\max}$ than an uninhibited reaction (choice B is wrong). Lastly, it can be seen on the graph that Curve 2 has a reduced $V_{\max}$ compared to Curve 3 (choice C is wrong).

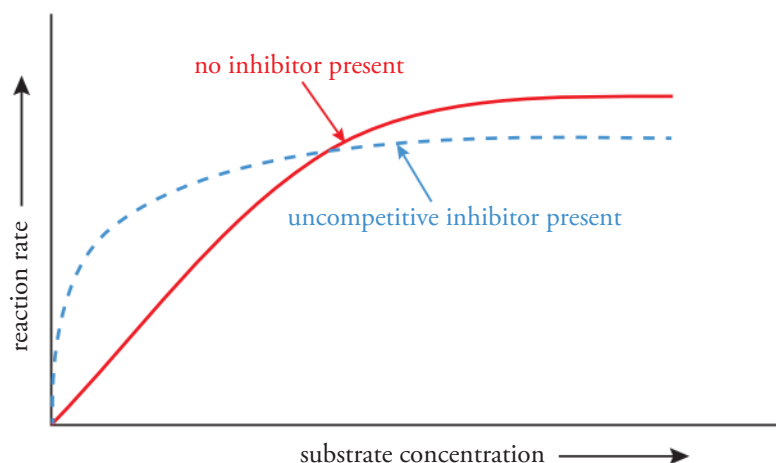


Figure 21 Uncompetitive inhibition

Mixed-type inhibition occurs when an inhibitor can bind to either the unoccupied enzyme or the enzyme-substrate complex. If the enzyme has greater affinity for the inhibitor in its free form, the enzyme will have a lower affinity for the substrate similar to competitive inhibition (K_m increases). If the enzyme-substrate complex has greater affinity for the inhibitor, the enzyme will have an apparently greater affinity (K_m decreases) for the substrate similar to what we saw in uncompetitive inhibition. On the rare occasion when it displays equal affinity in both forms, it would actually be a noncompetitive inhibitor (many textbooks list noncompetitive inhibition as an example of mixed-type inhibition). In each of these situations, the inhibitor binds to an allosteric site and additional substrate cannot overcome inhibition (V_{max} decreases).

4.8

Table 3 Changes in the Apparent V_{max} and K_m in Response to Various Types of Inhibition

Inhibition Type	$V_{max, app}$	$K_{m, app}$
Competitive	no change	↑
Noncompetitive	↓	no change
Uncompetitive	↓	↓
Mixed-type	↓	varies

4.9 LINEWEAVER-BURK PLOT

The Lineweaver-Burk plot is a graphical representation of enzyme kinetics using the Lineweaver-Burk Equation:

$$\frac{1}{V} = \left(\frac{K_m}{V_{\max}} \right) \left(\frac{1}{[S]} \right) + \frac{1}{V_{\max}}$$

The equation may appear intimidating but can be interpreted as a simple linear equation of the form: $y = mx + b$. The graph is called a double reciprocal plot because the y -axis is the inverse of the reaction rate $\left(\frac{1}{V} \right)$ and the x -axis is the inverse of the substrate concentration $\left(\frac{1}{[S]} \right)$. The key aspects you need to know about the Lineweaver-Burk plot are the following:

- 1) The slope of the graph is $\frac{K_m}{V_{\max}}$.
- 2) The y -intercept of the graph is $\frac{1}{V_{\max}}$.
- 3) The x -intercept of the graph is $-\frac{1}{K_m}$.

Recall that increasing the substrate concentration ($[S]$) increases the reaction rate V up to a point. An increase in substrate concentration, however, is a *decrease* in the inverse of the substrate concentration $\left(\frac{1}{[S]} \right)$. Thus, an interesting aspect of the Lineweaver-Burk plot is that an increase in substrate concentration means a decrease in the value along the x -axis. Similarly, an increase in the reaction rate V is a *decrease* in the inverse of the reaction rate $\left(\frac{1}{V} \right)$. Thus, as the reaction rate increases, the value along the y -axis decreases.

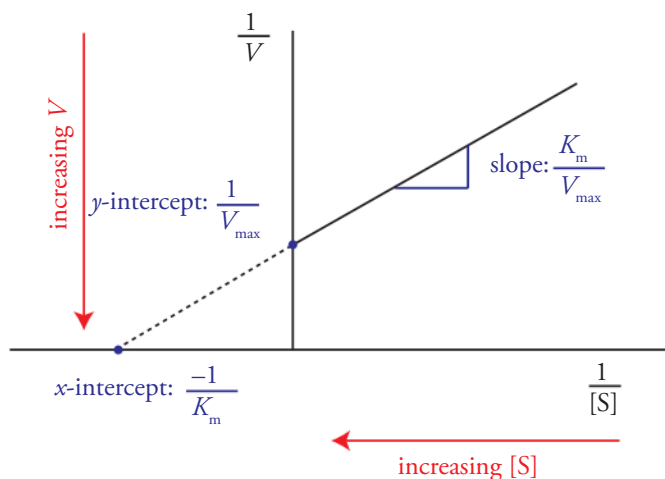


Figure 22 Lineweaver-Burk plot

- How would the Lineweaver-Burk plot change when a noncompetitive inhibitor is added?⁴³

⁴³ A noncompetitive inhibitor does not affect the K_m (so the x -intercept is unchanged) but does decrease the $V_{\max}$ (so the y -intercept increases and the slope increases).

Chapter 4 Summary

- Amino acids [AAs] consist of a tetrahedral α -carbon connected to an amino group, a carboxyl group, and a variable R-group, which determines the AA's properties.
- The isoelectric point of an AA is the pH at which the net charge on the molecule is zero; this structure is referred to as the zwitterion.
- Proteins consist of amino acids linked by peptide bonds, which are very stable. The primary structure of a protein consists of its amino acid sequence.
- The secondary structure of proteins (α -helices and β -sheets) is formed through hydrogen bonding interactions between atoms in the backbone of the molecule.
- The most stable tertiary protein structure generally places polar AAs on the exterior and nonpolar AAs on the interior of the protein. This minimizes interactions between nonpolar AAs and water, while optimizing interactions between side chains inside the protein.
- Proteins have a variety of functions in the body including [but not limited to] enzymes, structural roles, hormones, receptors, channels, antibodies, and transporters.
- Enzymes are biological catalysts that increase the rate of a reaction by lowering the activation energy.
- Unfavorable reactions in the cell are performed by coupling them to favorable reactions [such as ATP hydrolysis].
- Enzyme activity can be controlled via covalent modification, proteolytic cleavage, associations, or allosteric regulation.
- Competitive inhibitors bind at the active site of an enzyme, do not affect $V_{\max}$ but increase K_m .
- Noncompetitive inhibitors bind at an allosteric site of an enzyme, decrease $V_{\max}$ but do not change K_m .

- Uncompetitive inhibitors bind to the enzyme-substrate complex and reduce both K_m and V_{max} .
- Mixed-type inhibitors can bind to either the enzyme alone or the enzyme-substrate complex. They reduce V_{max} but have variable effects on K_m .
- The Lineweaver-Burk plot is a linear graph used to extrapolate the V_{max} and K_m of an enzyme. It is a visual aid that can be used to identify inhibitor type.

CHAPTER 4 FREESTANDING PRACTICE QUESTIONS

1. If a mutation changed an Arg residue in the protein interior to a Leu residue, what would be the likely effect on the ΔG of protein folding?
 - A. ΔG would become more positive.
 - B. ΔG would become more negative.
 - C. ΔG would remain unchanged.
 - D. The effect on ΔG cannot be determined without additional information.
2. Some inhibitors bind irreversibly to enzymes by covalent attachment. Would the kinetics seen under these conditions (V vs. $[S]$ curve) be similar to those seen with a reversible noncompetitive inhibitor?
 - A. Yes, because it would reduce the K_m of the reaction.
 - B. Yes, because the net effect would be a loss of active enzyme available for the reaction.
 - C. No, because if enough substrate binds to the active site, the reaction will reach V_{max} .
 - D. No, because K_m will increase and V_{max} will stay the same, similar to competitive inhibition.
3. Some enzymes can modify their substrate, and by this means, regulate its activity. In many instances, these modifications are not permanent since other enzymes can reverse them. Which of the following category of enzymes will irreversibly modify their substrate?
 - A. A kinase
 - B. A protease
 - C. A phosphatase
 - D. An acetylase
4. Which of the following would have the LEAST effect on the ability of an enzyme to bind its substrate?
 - A. Placing an enzyme that optimally functions at pH 7 in a pH 4 solution
 - B. Increasing the temperature of the enzyme's surroundings
 - C. Mutating the Glu and Asp residues in the active site to Lys and His
 - D. Changing the substrate from a tripeptide to a disaccharide
5. For a given enzyme concentration at a low substrate concentration, how does reaction rate change as the substrate concentration increases?
 - A. Logarithmically
 - B. Linearly
 - C. Exponentially
 - D. Indirectly
6. Both hemoglobin and myoglobin are proteins that carry oxygen in the human body. Hemoglobin exhibits cooperativity and is found in red blood cells, whereas myoglobin is not cooperative and is found in muscle cells. Which of the following is likely true regarding these oxygen-carrying proteins?
 - A. The O_2 saturation curve will be sigmoidal for myoglobin, but not hemoglobin.
 - B. Hemoglobin has a lower binding affinity for oxygen than myoglobin.
 - C. Hemoglobin is used to tightly bind oxygen in the body, while myoglobin is used to deliver oxygen to body cells.
 - D. Hemoglobin most likely consists of multiple protein subunits, whereas myoglobin may or may not consist of multiple protein subunits.
7. Which of the following will be true of the Lineweaver-Burk plot of an enzyme in the presence of increasing concentrations of a competitive inhibitor?
 - A. It will be a series of lines that intersect along the y-axis.
 - B. It will be a series of lines that intersect along the x-axis.
 - C. It will be a series of parallel lines.
 - D. It will be a series of lines that will intersect, but neither on the y- or x-axis.

CHAPTER 4 PRACTICE PASSAGE

The Michaelis-Menten equation expresses the relationship between reaction velocity (V) and substrate concentration ($[S]$) for enzymatic reactions involving a single substrate.

$$V = \frac{V_{\max}[S]}{K_m + [S]}$$

Equation 1

Equation 1 yields a hyperbolic curve when V is plotted against $[S]$ (Figure 1). Prior to the availability of computer programs that were able to perform non-linear regression, it was difficult to curve fit V vs. $[S]$ data accurately to obtain values for K_m and $V_{\max}$. One of the most commonly employed strategies to display the Michaelis-Menten equation in linear form was the Lineweaver-Burk formulation (Equation 2), in which taking the reciprocal of Equation 1 yields a relationship that can be represented by a linear plot of $1/V$ vs. $1/[S]$ (Figure 1):

$$\frac{1}{V} = \left(\frac{K_m}{V_{\max}} \right) \times \frac{1}{[S]} + \frac{1}{V_{\max}}$$

Equation 2

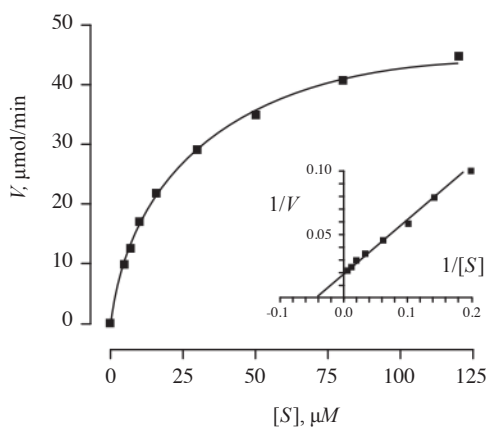


Figure 1 Sample V vs. $[S]$ plot with Lineweaver-Burk representation (inset)

Despite the advantages of this linear representation of enzymatic rate data, the Lineweaver-Burk plot introduces greater error when used to determine $V_{\max}$ and K_m . Thus, it is not recommended when highly accurate values of these kinetic parameters are desired. Instead, the continued utility of the Lineweaver-Burk plot lies in its ability to clearly differentiate modes of enzyme inhibition.

A researcher tested the ability of two different inhibitors to affect the activity of NADH oxidoreductase at two different concentrations. The results are displayed as Lineweaver-Burk plots (Figure 2).

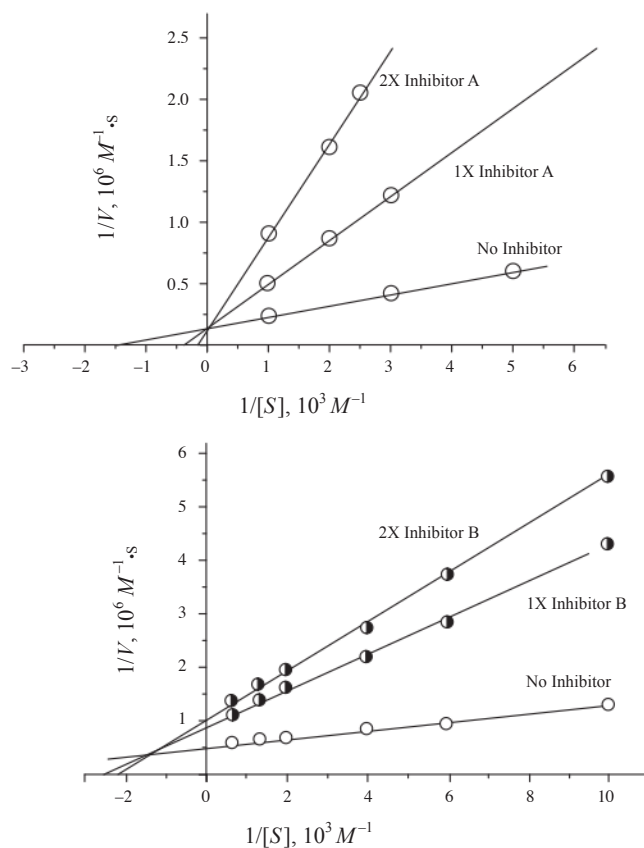


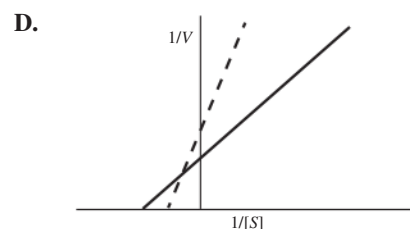
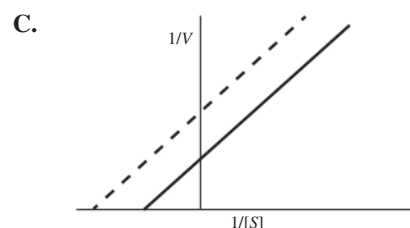
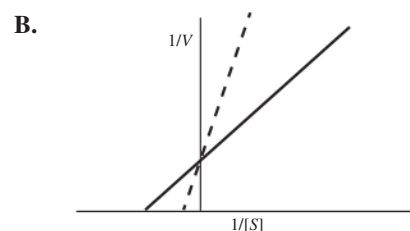
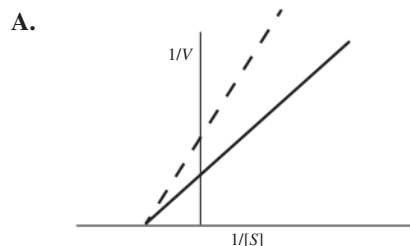
Figure 2 Inhibition of NADH oxidoreductase by two different inhibitors

Adapted from Saratovskikh, EA. Kinetics and Mechanism of Inhibition of Oxidation Enzymes by Herbicides. In *Herbicides, Advances in Research*. Edited by Price A and Kelton J. 2013

- Which of the following is the best estimate for the $V_{\max}$ of the enzyme represented by Figure 1?
 - 0.02 $\mu\text{mol} / \text{min}$
 - 23 $\mu\text{mol} / \text{min}$
 - 44 $\mu\text{mol} / \text{min}$
 - 52 $\mu\text{mol} / \text{min}$

2. Which of the following represent possible sources of error when estimating K_m and V_{max} from a Lineweaver-Burk plot?
- The inability to obtain negative values of $1/[S]$ often requires extrapolation of the linear fit over a long stretch lacking data points in order to determine the x -intercept.
 - Errors in the data obtained at low substrate concentration have a disproportionate impact on the determination of the best linear fit.
 - Visual estimation of K_m and V_{max} is more challenging from a linear plot as compared to the hyperbolic plot of V vs. $[S]$.
- A. I only
B. II only
C. I and II only
D. I, II, and III
3. Which of the following is true concerning the slope of a Lineweaver-Burk plot in the presence of a competitive versus a noncompetitive inhibitor?
- A. The slope changes in the presence of both a competitive and noncompetitive inhibitor.
B. The slope changes in the presence of a competitive inhibitor only.
C. The slope changes in the presence of a noncompetitive inhibitor only.
D. The slope does not change in the presence of either type of inhibitor.
4. All of the following are true concerning Inhibitor A EXCEPT:
- A. its effect on enzyme activity can be overcome by increasing the substrate concentration.
B. its effect on enzyme activity is equivalent to that of decreasing the concentration of the uninhibited enzyme.
C. it is unable to bind to the enzyme at the same time that substrate is bound.
D. its effect on enzyme activity is equivalent to that of decreasing the enzyme-substrate binding affinity.

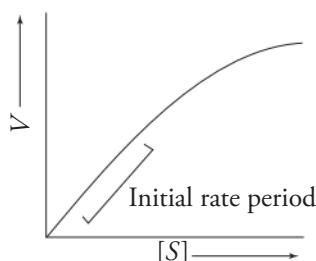
5. Which of the following represents the Lineweaver-Burk plot of an enzyme alone (solid line) and in the presence of a noncompetitive inhibitor (dashed line)?



SOLUTIONS TO CHAPTER 4 FREESTANDING PRACTICE QUESTIONS

1. **D** Although the hydrophobic effect is a major driving force in protein folding, the favorability of protein folding is also determined by the favorability of amino acid interactions. Generally, the positively charged Arg residue would not be favored in the nonpolar protein interior, relative to the nonpolar Leu residue. The Arg residue, however, may be favorably interacting with an acidic residue in the protein interior. As such, without additional information about other interactions the Arg is involved in, the effect on ΔG cannot be reasonably predicted (choices A, B, and C are wrong, and choice D is correct).
2. **B** An irreversible inhibitor (regardless of where it binds) will permanently deactivate some enzyme, reducing effective enzyme concentration. If the enzyme concentration is effectively lowered, $V_{\max}$ will be reduced (choices C and D are wrong). This is similar to what is seen in noncompetitive inhibition, in which the inhibitor binds to an allosteric site and turns the enzyme off. Even if the noncompetitive inhibitor is reversible, because at any given time some enzyme is “off,” the effective enzyme concentration is lowered and $V_{\max}$ is reduced. If K_m were affected, it would increase, not decrease; an increase in K_m indicates that the substrate-enzyme interaction has been compromised in some way (choice A is wrong).
3. **B** Proteases are enzymes that cleave their substrates at specific sites, permanently removing a part of the protein. This modification is practically irreversible—there are no enzymes that can reconnect proteins split by a protease. Choices A, C, and D are wrong because these categories of enzymes can reversibly modify their substrates. A kinase adds a phosphate to its substrate, but this modification can be reversed by a phosphatase that will remove the phosphate. An acetylase will add an acetyl group, while a deacetylase will remove it.
4. **B** Highlight the word “LEAST” and write “A B C D” on the notepad. Evaluate each answer choice, writing “Y” for big effect or “N” for a low effect; then choose the answer that stands out. Changing the pH of the solution to a non-optimal value, especially a pH that low, will affect the charge of the protein side chains and would have a big effect on enzyme structure and function (write “Y” next to “A” on the notepad). Increasing the temperature of an enzyme’s surroundings may or may not affect the likelihood of an enzyme binding its substrate. Increasing temperature will increase the kinetic energy of the substrate and make it more likely to enter the active site, unless the temperature is increased beyond the temperature of denaturation; since the question is not specific as to the magnitude of the temperature change, this is the choice that would likely have the LEAST effect (write “N” next to “B” on the notepad). Mutating acidic residues to basic residues will change the charge of the active site substantially and would likely affect the substrate’s ability to bind to the active site. This would have a big effect on enzyme function (write “Y” next to “C” on the notepad). Enzymes are very specific for their substrates; significantly altering the shape and size of the substrate by changing it from a tripeptide to a disaccharide will have a significant effect on enzyme function (write “Y” next to “D” on the notepad). Since choice B stands out with an “N” instead of a “Y,” it is the correct answer choice.

5. **B** At low substrate concentrations, the reaction rate increases linearly as the substrate concentration increases (see curve below). At or near saturation levels, the reaction rate begins to level off and does not change regardless of how much substrate is added. This is called the maximum velocity of reaction rate or $V_{\max}$.



6. **D** In order to exhibit cooperativity, a protein must be composed of multiple protein subunits that are able to interact with one another (or, less likely, it should have multiple active sites within the same protein); thus, it is likely that hemoglobin is composed of multiple subunits. Since myoglobin does not exhibit cooperativity, it may either be composed of one subunit or multiple non-cooperative subunits (choice D is correct). Cooperative enzymes usually exhibit sigmoidal curves; hemoglobin would be expected to exhibit a sigmoidal curve, but not myoglobin (choice A is wrong). Cooperativity is a phenomenon involving changes in binding affinity when a substrate is bound; thus, binding affinity between a cooperative protein and a non-cooperative protein is not directly comparable (choice B is wrong). Note, however, that at the typical oxygen concentration found in tissues (e.g., muscle), myoglobin would have to have the higher affinity in order to be able to “steal” oxygen from hemoglobin and store it. Since hemoglobin is found in red blood cells, it must be the delivery protein, while myoglobin in muscle cells is the storage protein (choice C is wrong).
7. **A** In a Lineweaver-Burk plot, the y -intercept represents $1/V_{\max}$, and the x -intercept represents $-1/K_m$. Because competitive inhibitors do not change the $V_{\max}$ and instead increase the K_m value, a Lineweaver-Burk plot with differing concentrations of competitive inhibitor will show intersecting lines that share a common y -intercept (choice A is correct). A common x -intercept would be indicative of an unchanged K_m value, as in the case of noncompetitive inhibition (choice B is wrong). A series of parallel lines is indicative of uncompetitive inhibition (choice C is wrong). A series of lines with any common intersection point not on the axes is indicative of mixed-type inhibition (choice D is wrong).

SOLUTIONS TO CHAPTER 4 PRACTICE PASSAGE

1. **D** This question highlights a main advantage of the Lineweaver-Burk plot. If Equation 2 is considered in $y = mx + b$ form, it is clear that the y -intercept (b) represents $1/V_{\max}$. Looking at Figure 1 in which $V_{\max}$ is approached asymptotically, choices A and B are clearly wrong. In particular, choice A is too small a number, and it represents the y -intercept of the Lineweaver-Burk plot, which is $1/V_{\max}$, not $V_{\max}$. Choices C and D both seem in reasonable range when looking at the hyperbolic graph, but estimation is difficult, as it has not completely leveled off at the highest shown substrate concentration. In the linear plot, this ambiguity is resolved as we are looking for the y -axis intercept. In this case, it occurs at just less than 0.02, so $V_{\max}$ is just greater than $1/0.02$ or 50 $\mu\text{mol}/\text{min}$.

2. **C** For Roman numeral questions, start by evaluating the item that appears in exactly two answer choices; whether it is true or false you can eliminate half the answers. In this case, none of the items appear exactly twice, so start with the shortest item, Item III. Item III is false: the passage suggests that the impetus for creating the Lineweaver-Burk formulation was the lack of computer programs that could accurately perform non-linear regression (fitting the hyperbola of a V vs. $[S]$ graph). Clearly, drawing a straight line through points is easier (choice D can be eliminated). Item II is the next shortest item, and it is true: data obtained at low substrate concentration (and thus higher $1/[S]$; in effect, the points at the upper right of the graph), which are likely to have more measurement error, have a larger role in determining the slope of the linear fit (choice A can be eliminated). Lastly, Item I is true: as it is impossible to have negative substrate concentrations, a Lineweaver-Burk plot will never have data points in negative x -axis ($1/[S]$) territory. As such, extrapolation of the line back to the x -axis will occur over a stretch of the graph lacking data points (choice B can be eliminated and choice C is correct).

3. **A** This question requires recognition of the meaning of the slope of a Lineweaver-Burk plot. Equation 2 is conveniently written in $y = mx + b$ format, in which the dependent variable (y) is $1/V$, the independent variable (x) is $1/[S]$, the y -intercept (b) is $1/V_{\max}$, and the slope (m) is $K_m/V_{\max}$. Competitive inhibition increases K_m , while non-competitive inhibition reduces $V_{\max}$. As both K_m and $V_{\max}$ contribute to the slope, both modes of inhibition will engender a change in slope of the Lineweaver-Burk plot.

4. **B** Make sure to highlight the word “EXCEPT” and write “A B C D” on the noteboard. Evaluate each answer choice, writing “T” if it is true or “F” if it is false, then choose the answer that stands out. If you evaluate the linear plot in Figure 2 in the presence of Inhibitor A, you can see that the y -intercept ($1/V_{\max}$) is not affected by the inhibitor, while the slope ($K_m/V_{\max}$) increases. Thus, the effect of this inhibitor is to increase K_m while leaving $V_{\max}$ unchanged. This is the signature of a competitive inhibitor. Because the substrate and inhibitor compete for active-site binding, increasing the concentration of the substrate will overcome the effect of the inhibitor (write “T” next to “A” on the noteboard). If enzyme concentration were decreased, $V_{\max}$ would also decrease; however, since this inhibitor does not affect $V_{\max}$, choice B is false (write “F” next to “B” on the noteboard). Since both the substrate and the inhibitor bind to the active site, they cannot both bind at the same time (write “T” next to “C” on the noteboard). As K_m is a measure of binding affinity of the enzyme for the substrate, the increase in K_m caused by this inhibitor is functionally equivalent to a decrease in affinity of the enzyme for its substrate (write “T” next to “D” on the noteboard). Since choice B stands out with an “F” instead of a “T,” it is the correct answer choice.

5. **A** A noncompetitive inhibitor binds to an allosteric site on the enzyme, effectively turning the enzyme off, but it does not affect the ability of the enzyme to bind substrate. Thus, $V_{\max}$ will be decreased and K_m will remain the same. On the Lineweaver-Burk plot, a decrease in $V_{\max}$ is represented by an increase in the y -intercept (if V decreases, $1/V$ increases; choice B can be eliminated). K_m is represented by the x -intercept; since K_m is the same, there should be no change in the x -intercept (choices C and D can be eliminated, and choice A is correct).